

Digital Integrated Circuit Design

HW-4: 9-Bit CLA Design

Problem:

Please design a 9-bit CLA based on the following specifications.

- The top module of 9-bit CLA is listed as follows,

```

module Hw4_9bitCLA(X,      // data input
                  Y,      // data input
                  Cin,     // Carry in
                  Sum,     // Sum output
                  Cout,    // Carry output
                  OV);     // Overflow

```

- The block diagram of 9-bit CLA is shown in Fig. 1. All of signals, **X** (the data input), **Y** (the data input), **S[8:0]** (the Sum output), **C9** (the carry output), and **OV** (**overflow**) have to be included in your design and testbench, **Hw4_CLA_rtb.v**.

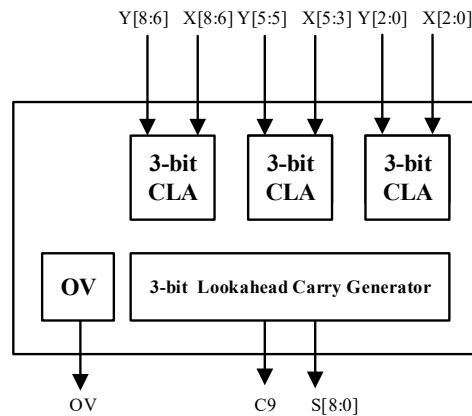


Fig. 1. Block diagram of 9-bit CLA.

- Importantly, your design has to detect the **overflow** and generate an alarm signal **OV**.
- The related timing diagrams are shown in Fig. 2 and 3 for reference.

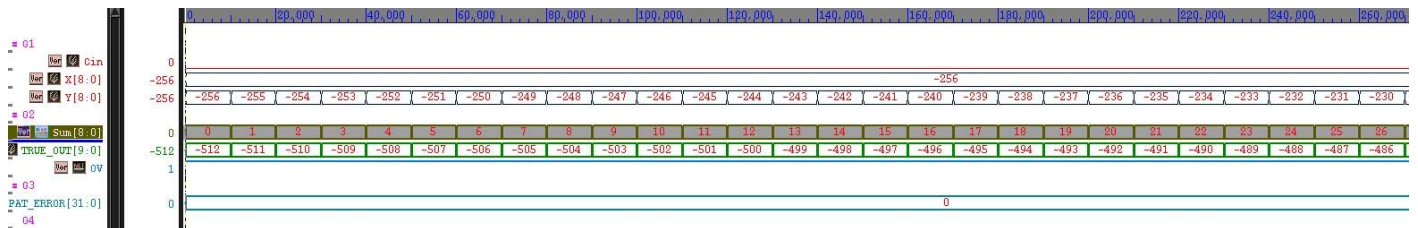


Fig. 2 A piece of timing diagram for overflow.



Fig. 3 A piece of timing diagram for the transition from overflow to non-overflow.